AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM271
Machine Learning and Deep Learning

Final Examination
Spring Semester 2025/2026

Date: 10/June /2026

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed: 3 Hours**

Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **NOT** allowed.

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PART – ONE Total Marks 10

Question No 1: Select the correct choice for each of the following statements.

1. Which component in a CNN is mainly used to reduce the spatial dimensions of feature maps?
   A) Convolution layer
   B) Pooling layer
   C) Dense layer
   D) Softmax layer
2. In an encoder-decoder model for machine translation, the encoder mainly:
   A) Generates the final translated sentence directly
   B) Converts the input sentence into a vector representation
   C) Removes all stop words from the sentence
   D) Performs clustering on the input text
3. Which NLP concept is used to convert word IDs into rich numerical representations?
   A) Max pooling
   B) Embedding
   C) Kernel trick
   D) Dropout
4. In a GAN, which network is responsible for judging whether a sample is real or generated?
   A) Encoder
   B) Generator
   C) Discriminator
   D) Recommender
5. Which recommender system approach uses the preferences of similar users to make suggestions?
   A) Content-based recommendation
   B) Collaborative filtering
   C) Knowledge-based recommendation
   D) Dimensionality reduction

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PART – TWO Total Marks 15

Question No 2: Differentiate between Autoencoders and GANs with suitable examples. **(6 Marks)**

Question No 3: Explain the attention mechanism in NLP and describe why it improves machine translation. **(5 Marks)**

Question No 4: What are recommender systems? Explain any three types of recommender systems. **(4 Marks)**

PART – THREE Total Marks 25

For the following questions, you are required to write the python code.

Scenario: Predicting customers' reviews

You are working as a Lead Data Scientist in a large e-commerce company. Your team wants to build an AI-based sentiment analysis system that can predict whether a customer review is positive (1) or negative (0).

You have been given a dataset named *"review_sentiment.csv"*. The dataset contains two columns:

- **review_text** → the customer review text
- **sentiment** → the target label (0 or 1)

Your task is to design and train a deep learning model that can learn from the text reviews and predict the sentiment.

Question No 5: Loading the Dataset **(3 Marks)**

Write Python code to load the dataset *"review_sentiment.csv"* into a DataFrame named *dataset*.

Question No 6: Dataset Splitting **(5 Marks)**

Write Python code to first separate the dataset into features (X) and target (y) variables, and then split the data into training and testing sets.

Question No 7: CNN Architecture Design **(12 Marks)**

Design a deep learning model for sentiment analysis using embedding and GRU/RNN layers. Include the necessary preprocessing steps.

Question No 8: Model Training and Prediction **(5 Marks)**

Write Python code to train the designed model and make predictions on the test dataset..

End of questions

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